

HTML TUTORIAL GUIDE

Beginner to Advanced



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July 5, 2026

Preface

This tutorial is designed to help beginners learn HTML from scratch. Each chapter builds upon the previous one and includes examples and explanations.

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1 Introduction to HTML

1.1 What is HTML?

HTML (**HyperText Markup Language**) is the standard markup language used to create and structure web pages. It provides the basic framework of a website by defining elements such as headings, paragraphs, images, links, tables, forms, and multimedia content.

HTML was created by **Sir Tim Berners-Lee** in 1991 to help researchers share information over the Internet. Today, HTML is one of the core technologies of the World Wide Web, alongside CSS and JavaScript.

The term **HyperText** refers to text that contains links to other documents or web pages. The term **Markup Language** refers to a system of tags used to describe the structure and presentation of content.

1.2 Why Learn HTML?

- Foundation of web development
- Easy to learn
- Works with CSS and JavaScript
- Required for creating websites

2 Basic HTML Tags

2.1 Understanding Basic HTML Tags

Before creating a web page, it is important to understand some basic HTML tags. Every HTML document starts with a few essential elements that help the browser understand and display the page correctly.

2.1.1 `<!DOCTYPE html>`

The `<!DOCTYPE html>` declaration tells the browser that the document is written in HTML5, which is the latest version of HTML.

Simple Explanation:

It is like telling a teacher which language you are writing an exam in. Before reading your answers, the teacher should know whether the paper is in English, Hindi, or Marathi. Similarly, the browser needs to know that the file is an HTML document.

Real-Life Example:

Imagine a book. The cover page tells you whether it is a Mathematics book, a Science book, or a History book. The `<!DOCTYPE html>` declaration acts like the cover page of a book.

Example:

```
<!DOCTYPE html>
```

2.1.2 `<head>`

The `<head>` section contains information about the web page that is useful for the browser but is usually not visible to visitors.

Simple Explanation:

It stores information such as the page title, links to CSS files, character encoding, and other settings.

Real-Life Example:

Think of a student. The student's name, roll number, and class are stored in the school records. Visitors do not see these records directly, but they help identify the student. The `<head>` section works in a similar way.

Example:

```
<head>
  <title>My Website</title>
</head>
```

The title appears on the browser tab.

2.1.3 `<body>`

The `<body>` section contains all the content that users can see on the web page.

Simple Explanation:

Everything displayed on the screen, such as headings, paragraphs, images, buttons, videos, and links, is placed inside the `<body>` tag.

Real-Life Example:

If a website is a house, the `<body>` is the rooms, furniture, television, and decorations that people actually see when they enter the house.

Example:

```
<body>
  <h1>Welcome</h1>
  <p>This is my first web page.</p>
</body>
```

2.1.4 Putting Everything Together

```
<!DOCTYPE html>

<html>
<head>
  <title>My First Website</title>
</head>
<body>
```

```
<h1>Welcome to HTML</h1>
<p>This is my first web page.</p>
</body>
</html>
```

Real-Life Analogy of a House

- `<!DOCTYPE html>` = The signboard that tells what type of building it is.
- `<head>` = The blueprint and documents of the house.
- `<body>` = The actual rooms and furniture that people see and use.

Together, these elements form the basic structure of every HTML web page.

3 HTML Headings and Paragraphs

3.0.1 <h1> to <h6> Heading Tags

Heading tags are used to define titles and subtitles on a web page. HTML provides six levels of headings, from <h1> to <h6>.

Simple Explanation:

Just as a book has a main title, chapter titles, and section titles, a web page uses heading tags to organize content.

- <h1> - Largest and most important heading
- <h2> - Second-level heading
- <h3> - Third-level heading
- <h4> - Fourth-level heading
- <h5> - Fifth-level heading
- <h6> - Smallest heading

Real-Life Example:

Consider a textbook:

- Book Title = <h1>
- Chapter Title = <h2>
- Section Title = <h3>
- Subsection Title = <h4> to <h6>

Example:

```
<h1>HTML Tutorial</h1>
<h2>Introduction</h2>
<h3>What is HTML?</h3>
```

Output:

HTML Tutorial (Largest Heading)

Introduction

What is HTML?

3.0.2 <p> Paragraph Tag

The paragraph tag is used to display a block of text on a web page.

Simple Explanation:

Whenever you want to write a sentence, description, article, or any text content, you place it inside the <p> tag.

Real-Life Example:

In a newspaper, the heading tells you the topic, while the paragraph provides the detailed information. Similarly, heading tags introduce a topic, and paragraph tags explain it.

Example:

```
<p>
HTML is the standard language used
to create web pages.
</p>
```

Output:

HTML is the standard language used to create web pages.

3.0.3 Combining Headings and Paragraphs

Headings and paragraphs are often used together to organize content.

```
<h1>My Website</h1>

<p>
Welcome to my website.
This site teaches HTML.
</p>

<h2>About HTML</h2>

<p>
```

```
HTML stands for HyperText Markup Language.  
It is used to create web pages.  
</p>
```

Real-Life Analogy

Think of a newspaper article:

- Headline = `<h1>`
- Section Titles = `<h2>` and `<h3>`
- Article Content = `<p>`

Without headings, readers may not know what the content is about. Without paragraphs, the information would be difficult to read. Therefore, headings and paragraphs work together to make web pages clear and well-organized.

4 Links and Images

4.0.1 <a> Link Tag

The <a> tag is used to create hyperlinks in HTML. A hyperlink allows users to navigate from one web page to another, open documents, or visit external websites.

Simple Explanation:

A link acts like a road sign that directs you to another destination when you click on it.

Real-Life Example:

Imagine you are in a shopping mall and see a sign that says "Food Court →". By following the sign, you reach the food court. Similarly, a hyperlink guides users to another page or website.

Syntax:

Example:

```
Visit Google </a>
```

Explanation:

- <a> is the anchor tag.
- href specifies the destination URL.
- The text between the tags is displayed as a clickable link.

Opening a Link in a New Tab

```
target="_blank">
Visit Google </a>
```

The `target="_blank"` attribute opens the link in a new browser tab.

4.0.2 Image Tag

The tag is used to display images on a web page.

Simple Explanation:

It allows you to add pictures, logos, icons, and graphics to make a website more attractive and informative.

Real-Life Example:

Consider a photo album. Without photos, the album would contain only text descriptions. The tag helps display actual images instead of just describing them.

Syntax:

Example:

```
alt="Beautiful Flower">
```

Explanation:

- src specifies the image location.
- alt provides alternative text if the image cannot be displayed.

Setting Image Width and Height

```
alt="Beautiful Flower"  
width="300"  
height="200">
```

This displays the image with a width of 300 pixels and a height of 200 pixels.

4.0.3 Using an Image as a Link

An image can also be made clickable by placing it inside a link tag.

```
alt="Google Logo"> </a>
```

When users click the image, they are redirected to the specified website.

Real-Life Analogy

- <a> (Link) = A road sign that directs you to a destination.
- (Image) = A photograph displayed in an album or on a wall.

Links help users navigate between pages, while images make web pages visually appealing and easier to understand.

5 Lists

5.0.1 Ordered List

An ordered list is used when the items need to appear in a specific sequence. The browser automatically numbers each item.

Simple Explanation:

Use an ordered list when the order of items is important.

Real-Life Example:

Consider the steps for making tea:

1. Boil water.
2. Add tea leaves.
3. Add milk and sugar.
4. Serve the tea.

The steps must be followed in order, so an ordered list is appropriate.

Syntax:

```
<ol>
  <li>First Item</li>
  <li>Second Item</li>
  <li>Third Item</li>
</ol>
```

Example:

```
<ol>
  <li>Wake Up</li>
  <li>Brush Teeth</li>
  <li>Go to College</li>
</ol>
```

5.0.2 Unordered List

An unordered list is used when the order of items does not matter. The browser displays bullet points instead of numbers.

Simple Explanation:

Use an unordered list when you simply want to group related items.

Real-Life Example:

A shopping list:

- Milk
- Bread
- Sugar
- Rice

The order of these items is not important.

Syntax:

```
<ul>
  <li>Item One</li>
  <li>Item Two</li>
  <li>Item Three</li>
</ul>
```

Example:

```
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

5.0.3 List Item Tag

Both ordered and unordered lists use the tag to define individual list items.

Example:

```
<li>HTML</li>
```

The tag represents one item in a list.

5.0.4 Real-Life Analogy

- Ordered List () = Instructions in a recipe or assembly manual.
- Unordered List () = Items on a shopping list.

Example Combining Both Lists

```
<h2>Web Development Roadmap</h2>

<ol>
  <li>Learn HTML</li>
  <li>Learn CSS</li>
  <li>Learn JavaScript</li>
</ol>

<h2>Technologies Used</h2>

<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

In this example, the roadmap uses an ordered list because the learning sequence matters, while the technologies are shown using an unordered list because their order is not important.

6 HTML Tables

6.0.1 <table> HTML Table

The <table> tag is used to display data in rows and columns. Tables help organize information in a structured format, making it easier to read and compare data.

Simple Explanation:

A table is like a grid consisting of rows and columns. It is commonly used to display schedules, student records, product information, and reports.

Real-Life Example:

Consider your college attendance register. It contains columns such as Roll Number, Name, and Attendance. This information is organized in a table format.

6.0.2 Basic Table Structure

HTML tables are created using the following tags:

- <table> - Creates the table.
- <tr> - Creates a table row.
- <th> - Creates a table heading cell.
- <td> - Creates a table data cell.

6.0.3 Syntax

```
<table>
  <tr>
    <th>Heading 1</th>
    <th>Heading 2</th>
  </tr>
```

```
<tr>
  <td>Data 1</td>
  <td>Data 2</td>
</tr>

</table>
```

6.0.4 Example: Student Information Table

```
<table border="1">

<tr>
  <th>Roll No</th>
  <th>Name</th>
  <th>Course</th>
</tr>

<tr>
  <td>101</td>
  <td>Rahul</td>
  <td>BCA</td>
</tr>

<tr>
  <td>102</td>
  <td>Priya</td>
  <td>BCA</td>
</tr>

</table>
```

Output:

Table 6.1: Student Information

Roll No	Name	Course
101	Rahul	BCA
102	Priya	BCA
103	Amit	BCA
104	Neha	BCA

6.0.5 Explanation of Table Tags

Tag	Purpose
<code><table></code>	Creates the main table structure.
<code><tr></code>	Creates a row in the table.
<code><th></code>	Creates a table heading cell. Text is bold and centered by default.
<code><td></code>	Creates a normal data cell containing table information.

6.0.6 Table with More Columns

```
<table border="1">

<tr>
  <th>Name</th>
  <th>Age</th>
  <th>City</th>
</tr>

<tr>
  <td>Amit</td>
  <td>20</td>
  <td>Mumbai</td>
</tr>

<tr>
  <td>Neha</td>
  <td>19</td>
  <td>Pune</td>
</tr>

</table>
```

6.0.7 Real-Life Analogy

Think of a classroom attendance sheet:

- The entire sheet = `<table>`
- Each horizontal line = `<tr>`
- Column titles (Name, Roll No) = `<th>`
- Student details = `<td>`

6.0.8 Advantages of Tables

- Organizes data neatly.
- Makes information easy to read.
- Helps compare multiple records.
- Commonly used for reports and schedules.

6.0.9 Summary

HTML tables are used to display structured data in rows and columns. The main tags used in creating tables are `<table>`, `<tr>`, `<th>`, and `<td>`. Tables are widely used in websites to present information such as student records, employee details, timetables, and product lists.

7 HTML Forms

7.1 HTML Forms

HTML forms are used to collect information from users. They allow visitors to enter data such as names, email addresses, passwords, phone numbers, feedback, and other details.

7.1.1 What is a Form?

A form is a section of a web page that contains input fields where users can enter information. The entered data can then be sent to a server for processing.

Simple Explanation:

A form is like an application form that you fill out when applying for a college, job, or event.

Real-Life Example:

When you register for a website, you usually enter:

- Name
- Email Address
- Password
- Phone Number

All these fields are part of an HTML form.

7.1.2 The <form> Tag

The <form> tag is used to create a form.

Syntax:


```
<form>
  Form Elements
</form>
```

7.1.3 Basic Form Example

```
<form>

<label>Name:</label> <input type="text">

<br><br>

<label>Email:</label> <input type="email">

<br><br>

<input type="submit">

</form>
```

7.1.4 Common Form Elements

Table 7.1: Common HTML Form Elements

Element	Purpose
<form>	Defines a form used to collect user information.
<input>	Creates input fields such as text boxes, passwords, radio buttons, and checkboxes.
<label>	Provides a descriptive label for an input field.
<textarea>	Allows users to enter multiple lines of text, such as comments or feedback.
<select>	Creates a drop-down list from which users can choose an option.
<option>	Defines an individual item within a drop-down list.
<button>	Creates a clickable button for actions such as submitting or resetting a form.

7.1.5 Input Types

The `type` attribute specifies the type of input field.

Table 7.2: Common HTML Input Types

Input Type	Purpose
<code>text</code>	Accepts single-line text input such as names and user-names.
<code>password</code>	Accepts passwords and hides the entered characters for security.
<code>email</code>	Accepts email addresses and performs basic email validation.
<code>number</code>	Accepts numeric values such as age, marks, or quantity.
<code>radio</code>	Allows the user to select only one option from a group of choices.
<code>checkbox</code>	Allows the user to select one or more options from a list.
<code>date</code>	Displays a date picker for selecting a date.
<code>submit</code>	Creates a button that submits the form data to the server.

7.1.6 Text Input Example

This field allows users to enter text such as their name.

7.1.7 Password Input Example

The entered characters are hidden for security.

7.1.8 Radio Button Example

```
<p>Gender:</p>

<input type="radio" name="gender">
Male

<input type="radio" name="gender">
Female
```

Radio buttons allow users to select only one option.

7.1.9 Checkbox Example

```
<p>Skills:</p>  
  
<input type="checkbox"> HTML <input type="checkbox"> CSS <input  
  type="checkbox"> JavaScript
```

Checkboxes allow users to select multiple options.

7.1.10 Drop-Down List Example

A drop-down list allows users to choose one option from many choices.

7.1.11 Text Area Example

```
<textarea rows="4" cols="30">  
</textarea>
```

A text area is used for longer messages such as comments and feedback.

7.1.12 Submit Button Example

The submit button sends the form data for processing.

7.1.13 Complete Registration Form

```
<form>  
  
<label>Name:</label> <input type="text">  
  
<br><br>  
  
<label>Email:</label> <input type="email">  
  
<br><br>
```

```
<label>Password:</label> <input type="password">  
  
<br><br>  
<input type="submit" value="Register">  
</form>
```

7.1.14 Real-Life Analogy

Think of a college admission form:

- Form = Admission Form
- Text Box = Name Field
- Radio Button = Gender Selection
- Checkbox = Hobbies Selection
- Drop-Down List = Course Selection
- Submit Button = Submit Application

7.1.15 Summary

HTML forms are used to collect information from users. The `<form>` tag acts as a container for input fields such as text boxes, radio buttons, checkboxes, drop-down lists, and buttons. Forms are essential for registration pages, login pages, surveys, contact forms, and online applications.

8 Advanced HTML Tags

8.1 Semantic Elements, Audio and Video in HTML

HTML5 introduced semantic elements and multimedia tags that make web pages more meaningful, accessible, and interactive.

8.1.1 Semantic Elements

Semantic elements clearly describe their meaning to both the browser and the developer. They make the structure of a web page easier to understand.

Simple Explanation:

Instead of using generic tags such as `<div>` everywhere, semantic tags tell us the purpose of each section of the webpage.

Real-Life Example:

Consider a newspaper:

- Newspaper Name = Header
- Menu = Navigation
- Main Article = Article
- Different Topics = Sections
- Contact Information = Footer

HTML semantic elements work in a similar way.

8.1.2 Example of Semantic Elements

```
<header>
  <h1>My Website</h1>
```

Tag	Purpose
<code><header></code>	Defines the introductory or top section of a webpage, usually containing the website title, logo, or heading.
<code><nav></code>	Contains navigation menus and links that help users move between pages or

```
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
</nav>

<section>
  <h2>HTML Tutorial</h2>

  ''
<article>
  <p>Learn HTML from basics.</p>
</article>
''

</section>

<footer>
  Copyright 2026
</footer>
```

8.1.3 Advantages of Semantic Elements

- Improve readability of code.
- Help search engines understand content.
- Improve website accessibility.
- Make website maintenance easier.

8.2 HTML Audio

The `<audio>` tag is used to embed sound files in a web page.

Simple Explanation:

It allows users to play music, podcasts, recorded lectures, and other audio files directly from the browser.

Real-Life Example:

When you listen to a song on a website without downloading it, the website is often using the `<audio>` tag.

8.2.1 Audio Syntax

```
type="audio/mpeg"> </audio>
```

8.2.2 Explanation

- `<audio>` creates an audio player.
- controls displays play, pause, and volume controls.
- `<source>` specifies the audio file.

8.2.3 Audio Example

```
type="audio/mpeg"> </audio>
```

This code creates an audio player that allows users to play the file `welcome.mp3`.

8.3 HTML Video

The `<video>` tag is used to display videos on a web page.

Simple Explanation:

It allows users to watch videos directly in the browser without using external software.

Real-Life Example:

Educational websites often use videos to explain concepts. These videos are commonly embedded using the `<video>` tag.

8.3.1 Video Syntax

```
type="video/mp4"> </video>
```

8.3.2 Explanation

- `<video>` creates a video player.
- `width` specifies the width of the player.
- `controls` displays play, pause, and volume controls.
- `<source>` specifies the video file.

8.3.3 Video Example

```
type="video/mp4"> </video>
```

This code displays a video player for the file `tutorial.mp4`.

8.3.4 Comparison of Audio and Video Tags

Table 8.1: Comparison of HTML Audio and Video Elements

Feature	Audio	Video
Tag Used	<code><audio></code>	<code><video></code>
Content Type	Sound only	Sound and visual content
Controls Available	Play, Pause, Volume	Play, Pause, Volume, Full Screen
Common Uses	Music, Podcasts, Audio Books	Tutorials, Movies, Lectures, Demonstrations
File Formats	MP3, WAV, OGG	MP4, WebM, OGG
Screen Display	No visual display	Displays video on the screen

8.3.5 Summary

Semantic elements provide meaningful structure to web pages and improve readability, accessibility, and SEO. The `<audio>` tag allows websites to play sound files, while the `<video>` tag enables video playback directly in the browser. These HTML5 features help create modern and interactive websites.

9 Conclusion

Congratulations! You have learned the fundamentals of HTML. Continue practicing and move on to CSS and JavaScript to become a web developer.

References

- `HTML5 Documentation`
- `Web Development Tutorials`
- `Practice Projects`